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=====
=== Huffman Cryptographic Compressor Menu ===
=====
1. Encrypt & Compress File
2. Decrypt & Decompress File
3. Exit Program
Enter Choice (1-3): 2
Enter Path to Source File: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Answer.txt
Enter Path to Output Target: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Final.txt

Validating cryptographic signature and extracting data stream...
Process Complete!
-----
=== ARCHIVE EXTRACTION METRICS ===
[-] Encrypted Archive Size: 585 Bytes
[+] Restored File Size      : 797 Bytes
[*] Restored Data Growth   : +212 Bytes returned to original state.
-----
Restored uncompressed file at: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Final.txt

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=== Huffman Cryptographic Compressor Menu ===
=====
1. Encrypt & Compress File
2. Decrypt & Decompress File
3. Exit Program
Enter Choice (1-3): 3

Closing engine streams. Goodbye!

-----
Process exited after 125.2 seconds with return value 0
Press any key to continue . . .
```



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=====
                        WELCOME TO THE SYSTEM
=====
Create Account - Enter Username: Zunaisha
Enter Password (min 8 characters): aumc2025

Account successfully created for Zunaisha!

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=== Huffman Cryptographic Compressor Menu ===
=====
1. Encrypt & Compress File
2. Decrypt & Decompress File
3. Exit Program
Enter Choice (1-3): 1
Enter Path to Source File: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Test1.txt
Enter Path to Output Target: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Answer.txt

Executing structural parsing and secure compression operations...
Process Complete!
-----
=== MEMORY OPTIMIZATION METRICS ===
[-] Original File Size   : 797 Bytes
[+] Compressed File Size: 585 Bytes
[*] Storage Space Saved : 26.60%
-----
Data safely locked within: C:\Users\SURFACE\OneDrive\Desktop\OOP-Spring2026\OOP-FinalProject\Answer.txt
```